



Computer Lab - Current Lighting

This project illustrates the current lighting arrangement of the Computer Lab of Department of Electrical Engineering, University of Moratuwa

Object
Computer Lab,
Department of Electrical
Engineering,
Faculty of Engineering,
University of Moratuwa

Editor
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Preface

Notes on planning:

The energy consumption quantities do not take into account light scenes and their dimming levels.

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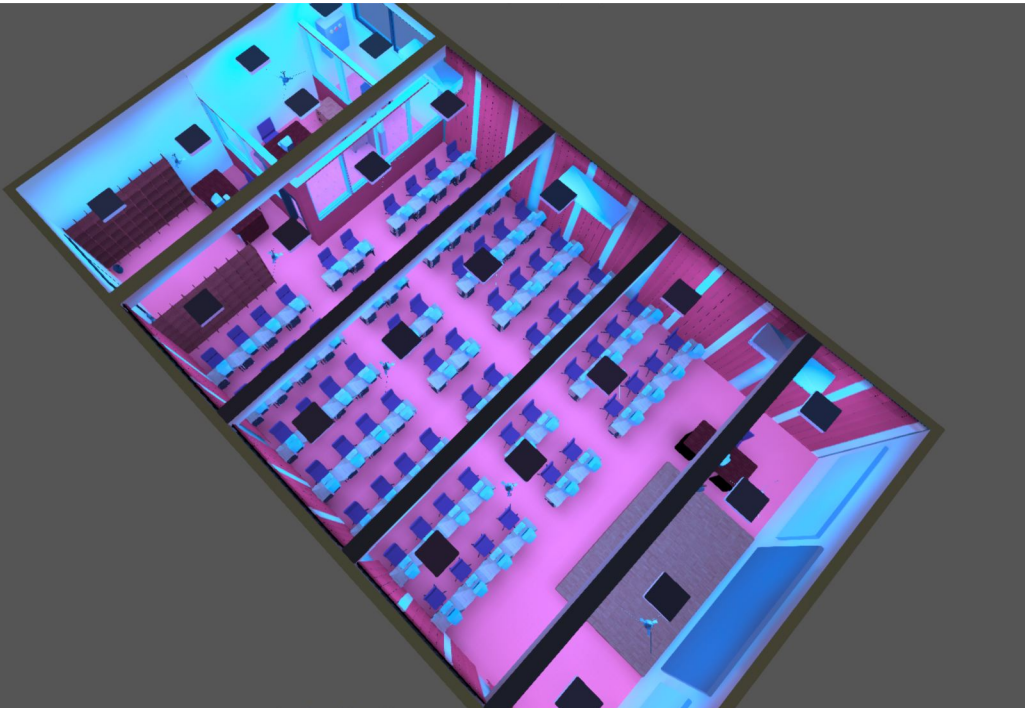
Contacts



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Description

This project illustrates the proposed lighting arrangement of the Computer Lab of Department of Electrical Engineering, University of Moratuwa. The project was implemented as an assignment given under module EE4410 - Electrical Services for Buildings.

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Luminaire list

 Φ_{total}

90112 lm

 P_{total}

748.0 W

Luminous efficacy

120.5 lm/W

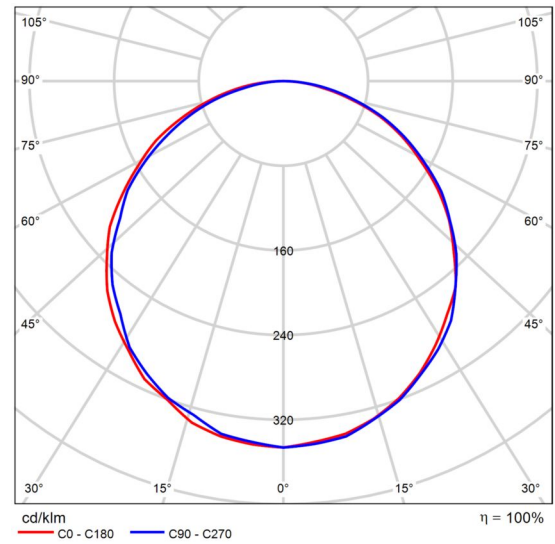
pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
22	Philips		RC048B+ LED41S PSU W60L60	34.0 W	4096 lm	120.5 lm/W

Product data sheet

Philips - RC048B+ LED41S PSU W60L60



P	34.0 W
Φ_{Lamp}	4100 lm
$\Phi_{Luminaire}$	4096 lm
η	99.90 %
Luminous efficacy	120.5 lm/W
CCT	3000 K
CRI	100



Polar LDC

Building 1

Luminaire list Φ_{total}

90112 lm

 P_{total}

748.0 W

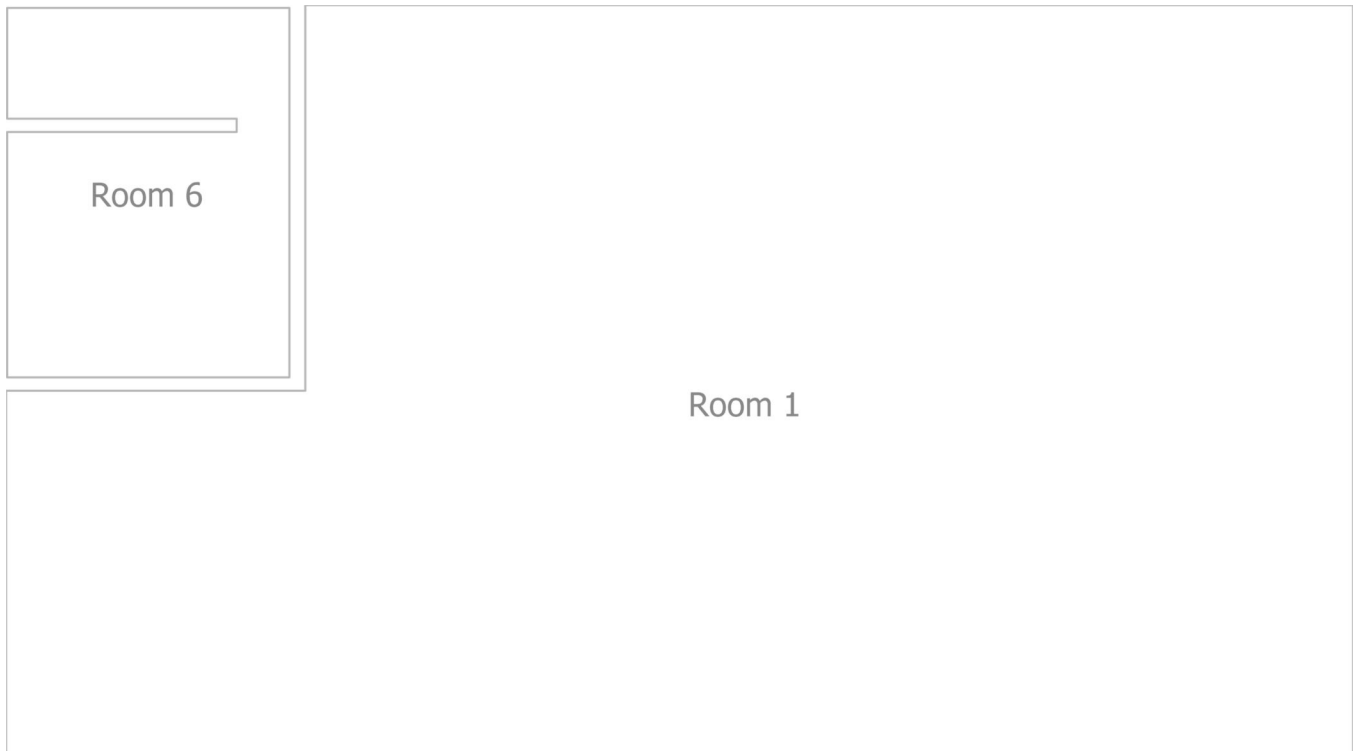
Luminous efficacy

120.5 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
22	Philips		RC048B+ LED41S PSU W60L60	34.0 W	4096 lm	120.5 lm/W

Building 1 · Storey 1 (Light scene 1)

Room list



Building 1 · Storey 1 (Light scene 1)

Room list

Room 1

P_{total} 612.0 W	A_{Room} 141.83 m ²	Lighting power density 4.31 W/m ² = 1.25 W/m ² /100 lx (Space) 5.26 W/m ² = 1.52 W/m ² /100 lx (Working plane)	$\bar{E}_{\text{perpendicular (Working plane)}}$ 346 lx
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pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
18	Philips		RC048B+ LED41S PSU W60L60	34.0 W	4096 lm

Room 6

P_{total} 136.0 W	A_{Room} 16.01 m ²	Lighting power density 8.50 W/m ² = 1.84 W/m ² /100 lx (Space) 11.77 W/m ² = 2.55 W/m ² /100 lx (Working plane)	$\bar{E}_{\text{perpendicular (Working plane)}}$ 461 lx
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pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
4	Philips		RC048B+ LED41S PSU W60L60	34.0 W	4096 lm

Building 1 · Storey 1

Luminaire list Φ_{total}

90112 lm

 P_{total}

748.0 W

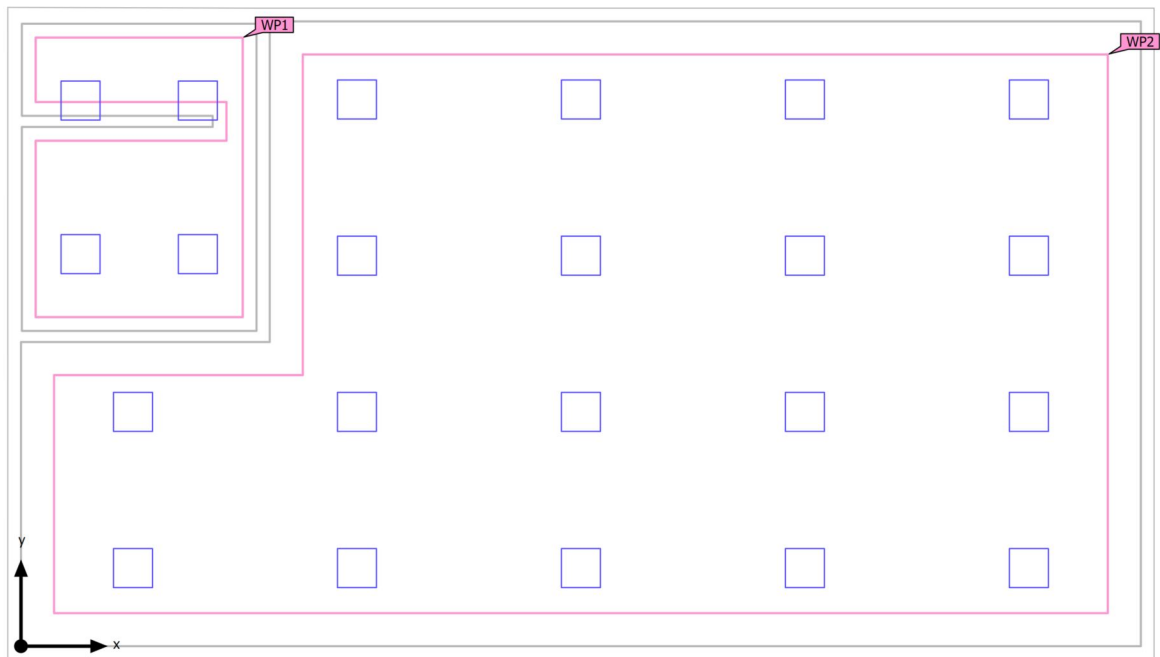
Luminous efficacy

120.5 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
22	Philips		RC048B+ LED41S PSU W60L60	34.0 W	4096 lm	120.5 lm/W

Building 1 · Storey 1 (Light scene 1)

Calculation objects



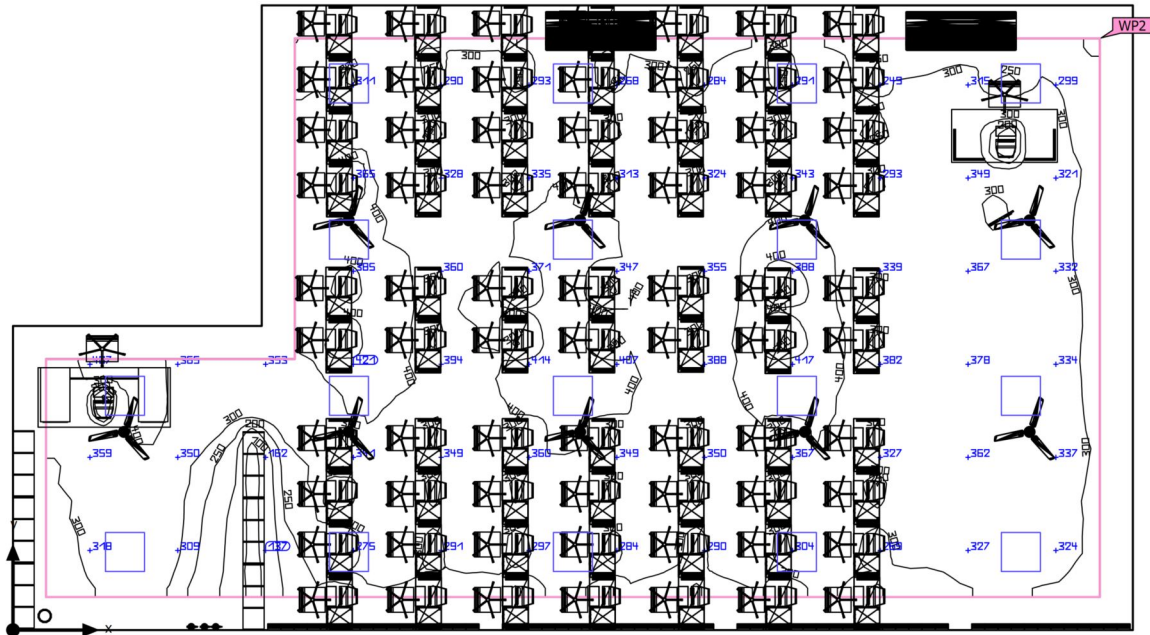
Building 1 · Storey 1 (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.209 m	461 lx (≥ 300 lx) ✓	336 lx	552 lx	0.73 (≥ 0.60) ✓	0.61	WP1
Working plane (Room 1) Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.500 m	346 lx (≥ 300 lx) ✓	69.6 lx	449 lx	0.20 (≥ 0.60) ✗	0.16	WP2

Building 1 · Storey 1 · Room 1 (Light scene 1)

Summary

Ground area	141.83 m ²		
Reflection factors	Ceiling: 60.5 %, Walls: 73.6 %, Floor: 28.8 %	Mounting height	3.658 m
Maintenance factor	0.80 (fixed)	Height _{Working plane}	0.800 m
		Wall zone _{Working plane}	0.500 m

Building 1 · Storey 1 · Room 1 (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	346 lx	≥ 300 lx	✓	WP2
	$U_o (g_1)$	0.20	≥ 0.60	✗	WP2
	Lighting power density	5.26 W/m ²	–		
		1.52 W/m ² /100 lx	–		
Energy estimation ⁽²⁾	Consumption	814 kWh/a	max. 5000 kWh/a	✓	
Space	Lighting power density	4.31 W/m ²	–		
		1.25 W/m ² /100 lx	–		

(1) Based on a rectangular space of 16.942 m x 9.449 m and SHR of 0.25.

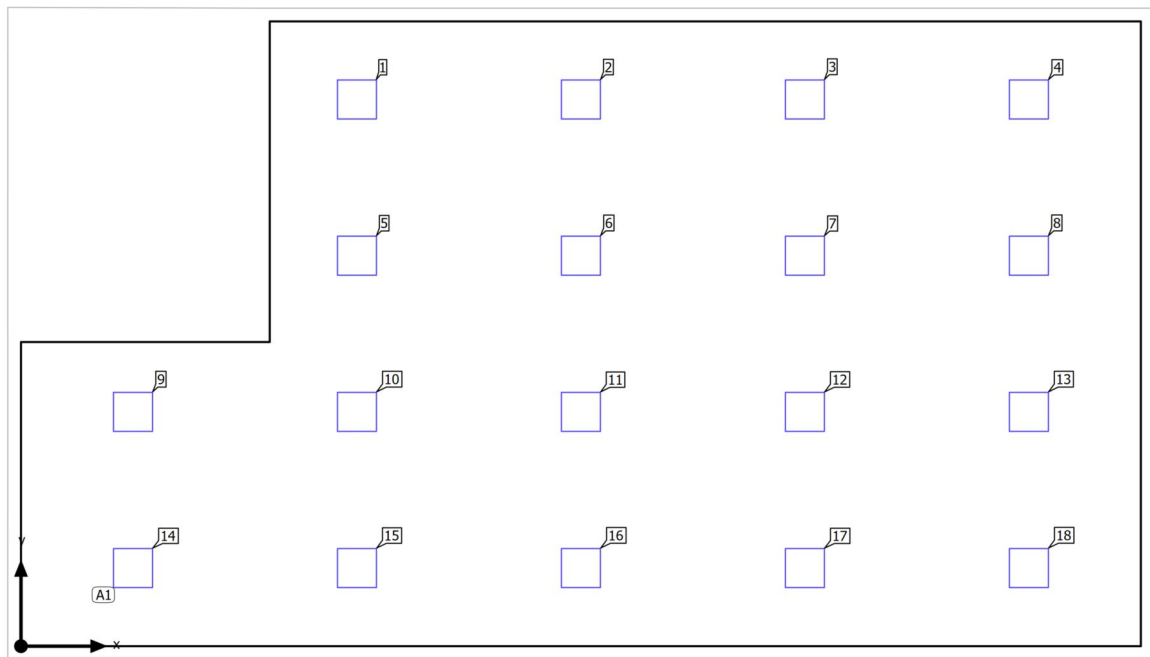
(2) Calculated using DIN:18599-4.

Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))

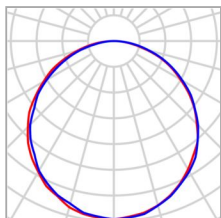
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R _{UG}	P	Φ	Luminous efficacy
18	Philips		RC048B+ LED41S PSU W60L60	–	34.0 W	4096 lm	120.5 lm/W

Building 1 · Storey 1 · Room 1

Luminaire layout plan

Building 1 · Storey 1 · Room 1

Luminaire layout plan

Manufacturer	Philips	P	34.0 W
Article name	RC048B+ LED41S PSU W60L60	Φ _{Luminaire}	4096 lm
Fitting	1x LED		

18 x Philips RC048B+ LED41S PSU W60L60

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	1.694 m / 1.181 m / 3.658 m	5.083 m	8.268 m	3.658 m	1
X-direction	5 pcs., Centre - centre, 3.388 m	8.471 m	8.268 m	3.658 m	2
Y-direction	4 pcs., Centre - centre, 2.362 m	11.859 m	8.268 m	3.658 m	3
Arrangement	A1	15.248 m	8.268 m	3.658 m	4
		5.083 m	5.906 m	3.658 m	5
		8.471 m	5.906 m	3.658 m	6
		11.859 m	5.906 m	3.658 m	7
		15.248 m	5.906 m	3.658 m	8
		1.694 m	3.543 m	3.658 m	9
		5.083 m	3.543 m	3.658 m	10
		8.471 m	3.543 m	3.658 m	11
		11.859 m	3.543 m	3.658 m	12
		15.248 m	3.543 m	3.658 m	13

Building 1 · Storey 1 · Room 1

Luminaire layout plan

X	Y	Mounting height	Luminaire
1.694 m	1.181 m	3.658 m	14
5.083 m	1.181 m	3.658 m	15
8.471 m	1.181 m	3.658 m	16
11.859 m	1.181 m	3.658 m	17
15.248 m	1.181 m	3.658 m	18

Building 1 · Storey 1 · Room 1

Luminaire list Φ_{total}

73728 lm

 P_{total}

612.0 W

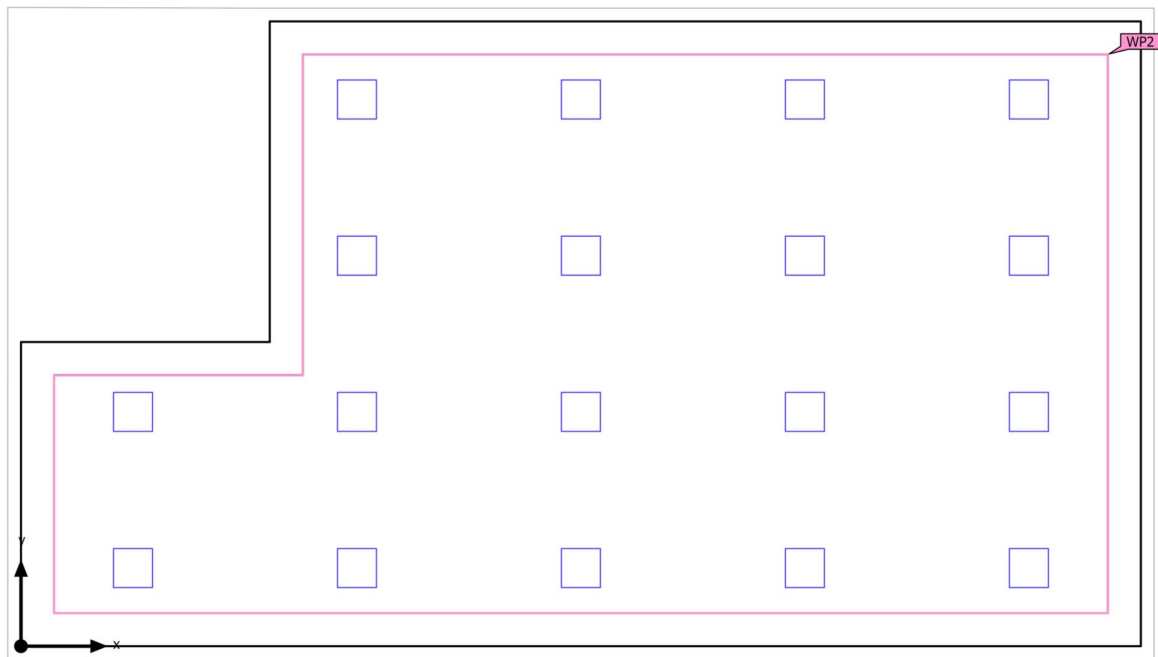
Luminous efficacy

120.5 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
18	Philips		RC048B+ LED41S PSU W60L60	34.0 W	4096 lm	120.5 lm/W

Building 1 · Storey 1 · Room 1 (Light scene 1)

Calculation objects



Building 1 · Storey 1 · Room 1 (Light scene 1)

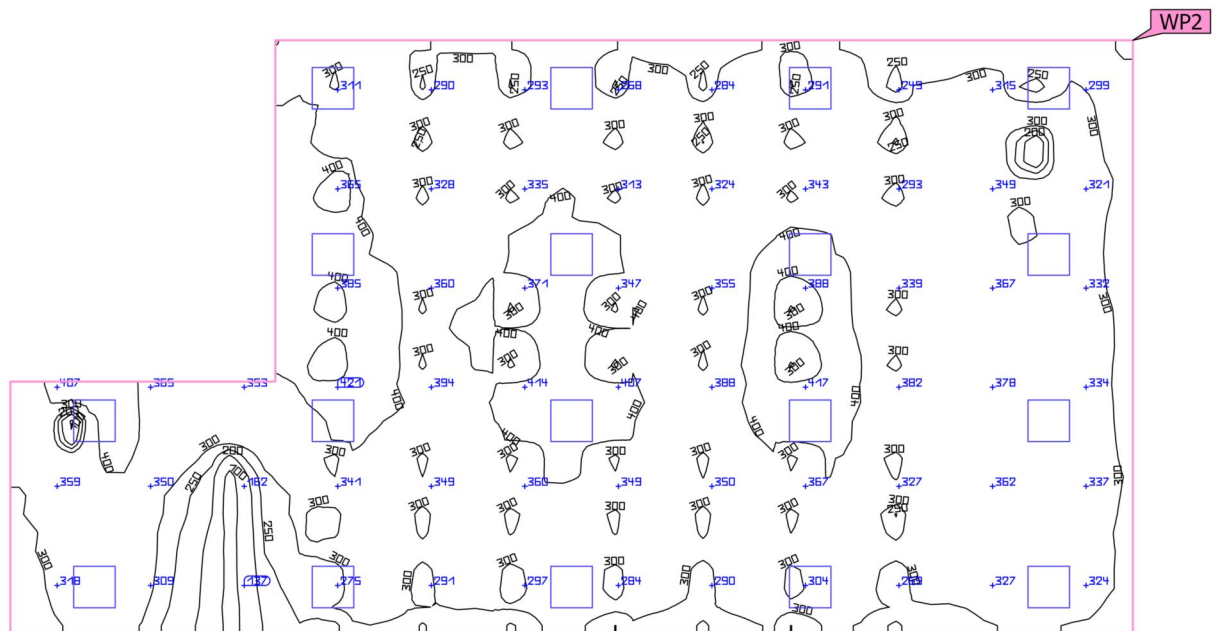
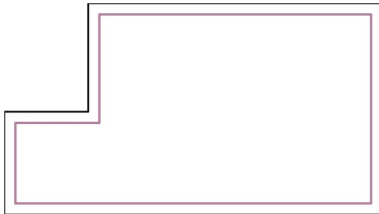
Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (Room 1) Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.500 m	346 lx (≥ 300 lx) ✓	69.6 lx	449 lx	0.20 (≥ 0.60) ✗	0.16	WP2

Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))

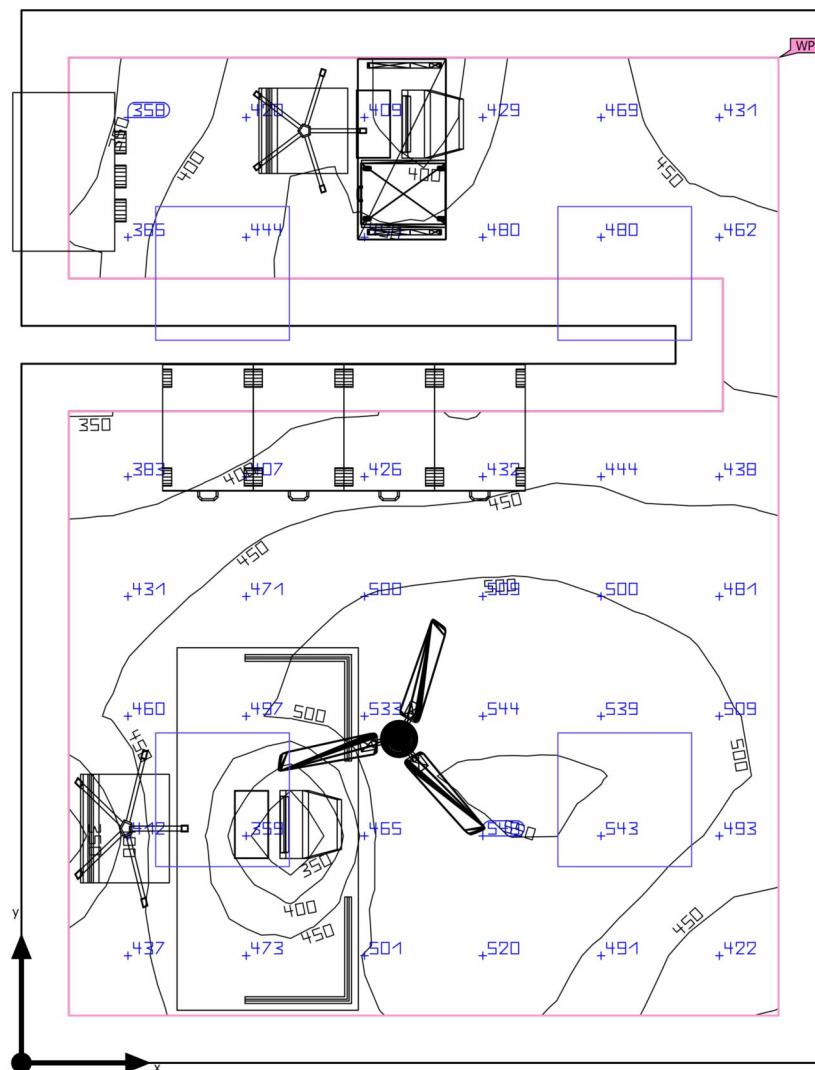
Building 1 · Storey 1 · Room 1 (Light scene 1)

Working plane (Room 1)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (Room 1)	346 lx	69.6 lx	449 lx	0.20	0.16	WP2
Perpendicular illuminance (adaptive)	(≥ 300 lx)			(≥ 0.60)		
Height: 0.800 m, Wall zone: 0.500 m	✓			✗		

Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))

Building 1 · Storey 1 · Room 6 (Light scene 1)

Summary

Ground area	16.01 m ²	Clearance height	3.658 m
Reflection factors	Ceiling: 72.5 %, Walls: 58.5 %, Floor: 23.0 %	Mounting height	3.658 m
Maintenance factor	0.80 (fixed)	Height _{Working plane}	0.800 m
		Wall zone _{Working plane}	0.209 m

Building 1 · Storey 1 · Room 6 (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	461 lx	≥ 300 lx	✓	WP1
	$U_o (g_1)$	0.73	≥ 0.60	✓	WP1
	Lighting power density	11.77 W/m ²	–		
		2.55 W/m ² /100 lx	–		
Energy estimation ⁽²⁾	Consumption	[109 - 181] kWh/a	max. 600 kWh/a	✓	
Space	Lighting power density	8.50 W/m ²	–		
		1.84 W/m ² /100 lx	–		

(1) Based on a rectangular space of 3.550 m x 4.645 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

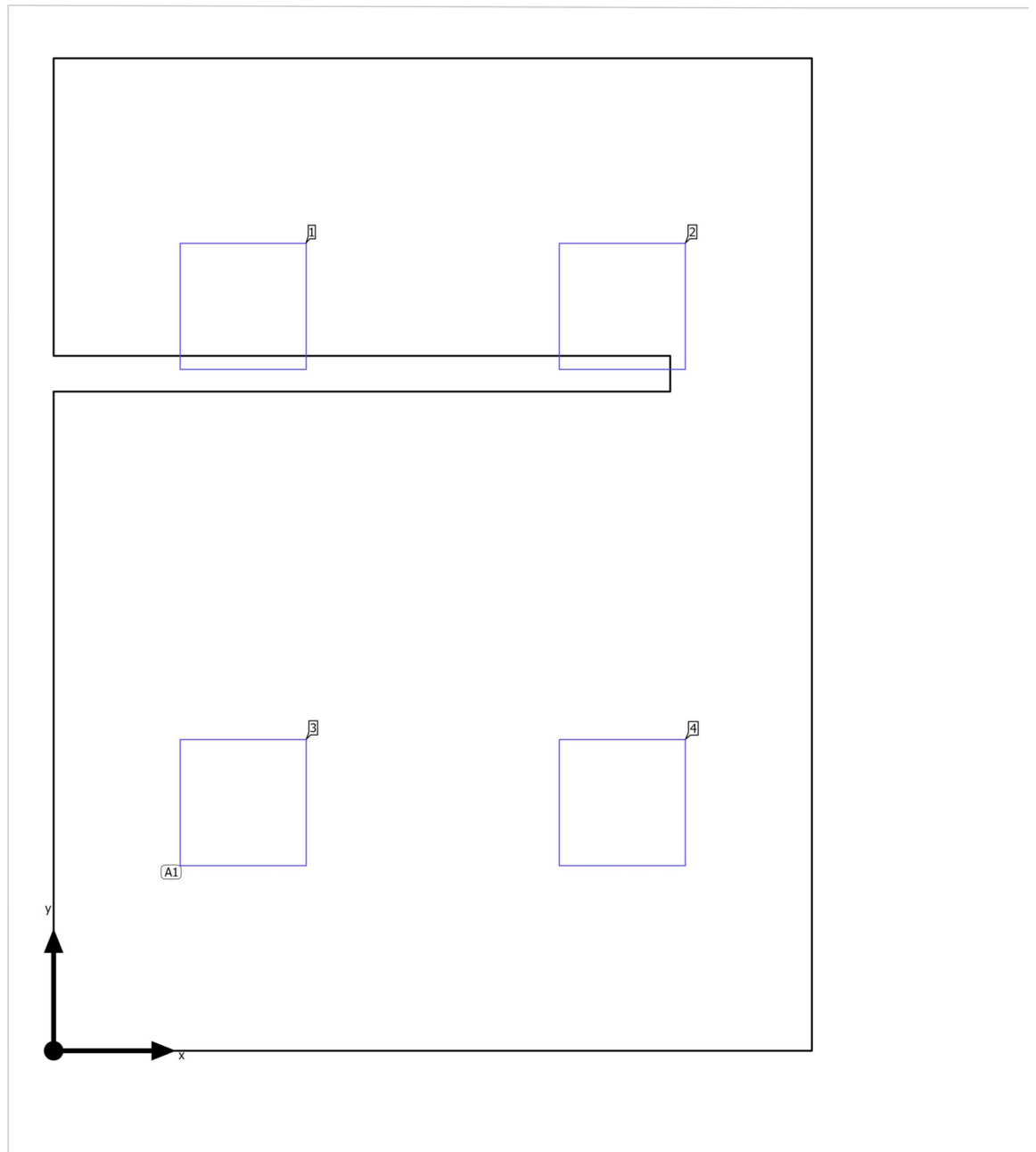
Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))

Luminaire list

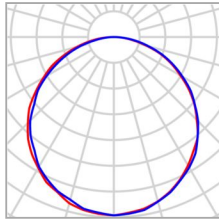
pcs.	Manufacturer	Article No.	Article name	R _{UG}	P	Φ	Luminous efficacy
4	Philips		RC048B+ LED41S PSU W60L60	–	34.0 W	4096 lm	120.5 lm/W

Building 1 · Storey 1 · Room 6

Luminaire layout plan



Building 1 · Storey 1 · Room 6

Luminaire layout plan

Manufacturer	Philips	P	34.0 W
Article name	RC048B+ LED41S PSU W60L60	$\Phi_{\text{Luminaire}}$	4096 lm
Fitting	1x LED		

4 x Philips RC048B+ LED41S PSU W60L60

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	0.888 m / 1.161 m / 3.658 m	0.888 m	3.484 m	3.658 m	1
X-direction	2 pcs., Centre - centre, 1.775 m	2.663 m	3.484 m	3.658 m	2
Y-direction	2 pcs., Centre - centre, 2.323 m	0.888 m	1.161 m	3.658 m	3
		2.663 m	1.161 m	3.658 m	4
Arrangement	A1				

Building 1 · Storey 1 · Room 6

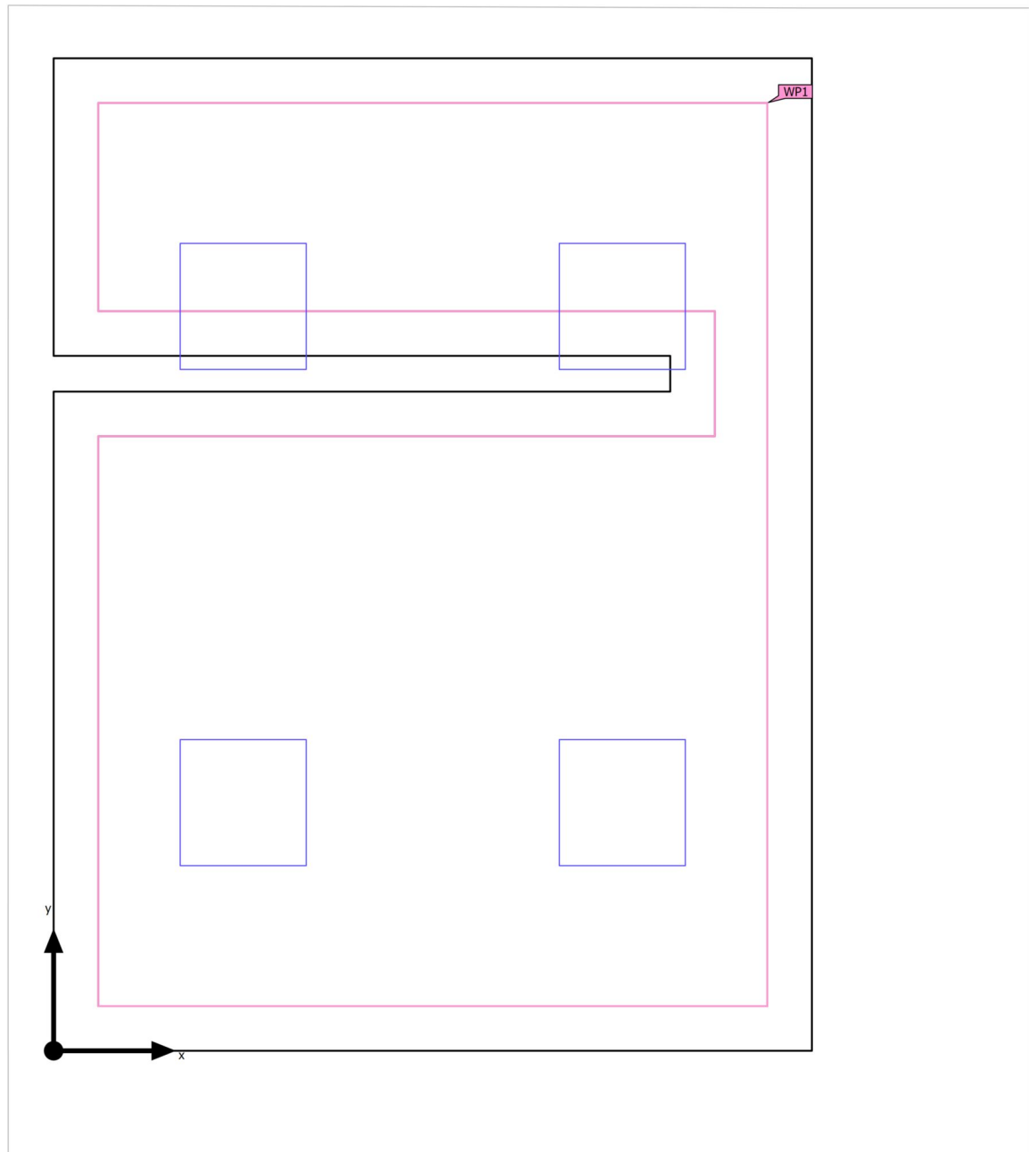
Luminaire list

Φ_{total} 16384 lm	P_{total} 136.0 W	Luminous efficacy 120.5 lm/W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
4	Philips		RC048B+ LED41S PSU W60L60	34.0 W	4096 lm	120.5 lm/W

Building 1 · Storey 1 · Room 6 (Light scene 1)

Calculation objects



Building 1 · Storey 1 · Room 6 (Light scene 1)

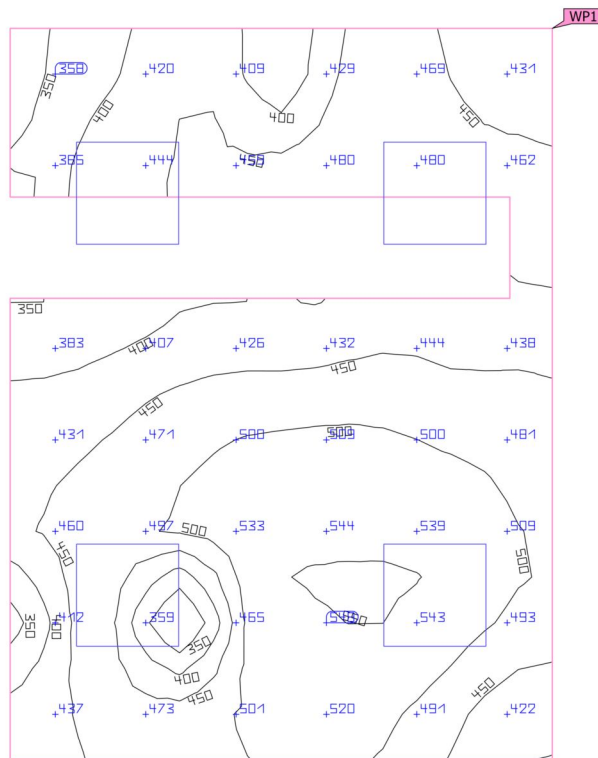
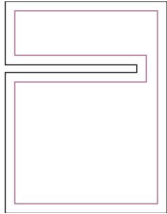
Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.209 m	461 lx (≥ 300 lx) ✓	336 lx	552 lx	0.73 (≥ 0.60) ✓	0.61	WP1

Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))

Building 1 · Storey 1 · Room 6 (Light scene 1)

Working plane

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane	461 lx	336 lx	552 lx	0.73	0.61	WP1
Perpendicular illuminance (adaptive)	≥ 300 lx			≥ 0.60		
Height: 0.800 m, Wall zone: 0.209 m	✓			✓		

Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))